

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Second Semester of M. Tech. Mechanical Engineering (CAD/CAM) Examination****May 2018****ME 772/ME 716.01 Industrial Tribology****Date: 10.05.2018, Thursday****Time: 01:30 p.m. to 04:30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q-1 (a)** Which are the two best surfaces that produce minimum friction levels? [02]
- (b)** Why does contact occur in a tribo-pair interaction, only at asperities? [02]
- (c)** Identify any two examples of wear and two examples of friction can be beneficial in day to day application. [02]
- (d)** What do you mean by viscosity index? [01]
- Q-2 (a)** Enumerate the different methods of studying the surface. Describe with neat sketch profilometer. [07]
- (b)** What is a better method for quantifying surface roughness, average roughness or root mean square roughness and why? [04]
- (c)** Discuss effect of pressure and temperature on viscosity of the lubricating oil. [03]

OR

- Q-2 (a)** Elaborate the working of tribometer available in lab for measurement of friction with neat sketches? [07]
- (b)** Classify the friction with applications. [04]
- (c)** Why static friction is higher than kinetic friction .Justify with proper logic. [03]
- Q-3 (a)** Which are the main criteria for classification of wear? Classify it. [04]
- (b)** What do you mean by abrasive and surface fatigue wear? [06]
- (c)** Elaborate the effect of various parameters on wear. [04]

OR

- Q-3 (a)** Explain the modified adhesion or junction growth theory of friction. [04]
- (b)** Explain the delamination theory of wear. State various assumptions. [05]
- (c)** What are the sources and causes of wear and effect of wear on properties of metals? [05]

SECTION – II

- Q-4 (a)** Clarify effect of adding additives to the lubricants? [02]
- (b)** What is fluid film lubrication? What is the difference between hydrostatic and hydrodynamic lubrication? [04]
- Q-5 (a)** State principle of hydrodynamic lubrication in journal bearing along with explanation. [03]
- (b)** What are silent features of good lubricants pertaining to lubricity? [04]
- (c)** Explain the mechanism of pressure development in oil film of hydrodynamic journal bearing. [07]

OR

- Q-5 (a)** Explain in brief the various arrangement of hydrostatic lubrication systems. [07]
- (b)** State advantage, limitation and applications of hydrostatic bearings. [07]
- Q-6** Write short note on **any three**. [15]
- (a)** Derive Reynold's equation for 3-D hydrodynamic lubrication. What are different assumptions made in this derivation?
- (b)** In which applications are gas lubricants used? Write down merits and limitations of gas lubricated bearings.
- (c)** Why according to the Stribeck curve hydrodynamic lubrication is better than boundary layerlubrication; even then in many applications we use boundary layer lubrication?
- (d)** Discuss different types of energy losses in hydrostatic bearings.
